RMweekly

Norah Jones

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1 Preface

This is a Quarto book. To learn more about Quarto books visit https://quarto.org/docs/books.

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2 Introduction

This is a book created from markdown and executable code.

See Knuth (1984) for additional discussion of literate programming.

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3 Summary

In summary, this book has no content whatsoever.

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4 Week1 — Getting started with remote sensing

4.1 1. Summary

"This week's content has shifted my perspective on remote sensing imagery from merely 'aesthetic images' to **quantifiable spectral data**. Different land covers—such as water, vegetation, and buildings—possess unique **spectral response characteristics**, exhibiting significant variations across different bands. Sensors, in essence, capture the **radiance or reflectance information** of the Earth's surface across the electromagnetic spectrum."

4.1.1 1.1 Four types of resolution

- **Spatial Resolution**: A smaller pixel size provides finer detail, whereas a larger pixel size is more prone to the mixed pixel effect.
- **Spectral Resolution**: Refers to the number and width of spectral bands. Narrower and more numerous bands allow for better differentiation of subtle spectral signatures between materials.
- **Temporal Resolution**: The frequency of repeat observations over the same area.
- **Radiometric Resolution** This resolution defines the sensor's sensitivity to changes in light intensity. By quantizing the received energy into finer brightness levels, higher resolution results in much smoother color or grayscale transitions.

4.1.2 1.2 Feature Identification

Utilizing QGIS and SNAP, I employed true-color/false-color composites and histogram stretching to rapidly identify the spatial patterns of water bodies, vegetation, and built-up areas. This process enabled a direct comparison of the differences in scale and level of detail between Sentinel and Landsat imagery. Beyond mere visual observation, this step involves a quality screening for issues such as cloud cover, shadows, stripping, and edge NoData values, providing a **basis** for defining AOIs and sampling strategies in subsequent stages.

4.2 2. Applications

4.2.1 2.1 Processing levels

The most common classification for mainstream remote sensing products is based on their processing levels. Specifically, NASA's Level 0–4 definitions serve as the widely adopted reference framework across the industry NASA Data Processing Levels Processing Levels.

Table 4.1: Remote Sensing Data Processing Levels

Level	Name	Key definitions
L0	Raw Data	Raw sensor streams, typically as original telemetry packets.
L1	Geometric Correction	Data is converted to physical units with Earth location information attached.
L2	Surface Physical Variables	Derived geophysical variables. The most critical step is Atmospheric Correction
L3	Spatio-temporal Composites	Mapped and gridded variables. Data is resampled onto a global or regional grid, often involving spatial mosaicking and temporal compositing.
L4	Higher-level Analysis Products	Integrated output from models.

4.2.2 3.1 Key Sensors

- **Sentinel-1 (SAR)**, with its cloud-penetrating and day-and-night imaging capabilities, is the preferred choice for event monitoring in cloudy or disaster-stricken scenarios.
- **Sentinel-2** is better suited for fine-grained surface interpretation, such as urban textures, detailed vegetation and water distribution, or short-term changes.
- **Landsat**'s long-term data record and stable calibration make it ideal for **lo**ng-term trend analysis, including decadal land-use changes and urban expansion.
- **MODIS and VIIRS** excel due to their high temporal resolution, capturing seasonal dynamics and regional indicators.

4.3 3. Reflection

I used to rely more on QGIS/SNAP because it is highly intuitive: by switching a band combination or applying a simple stretch, differences between land-cover types become visible

immediately. It is also easier to spot issues such as cloud and shadow contamination. However, this week I realised that as the data scale grows, the strengths of SNAP can turn into limitations: the same steps must be repeated manually, the workflow is harder to standardise, and it becomes more difficult to ensure consistent processing choices across runs. By contrast, code-based approaches feel more abstract at first, because they require translating what I do in the software into variables, parameters, and functions. Yet once the workflow is scripted, it can be run in batch, reused across datasets, and every processing decision is recorded. Going forward, an ideal strategy for me is to use the SNAP for rapid exploration and quality control, while relying on code for reproducible, scalable batch processing and quantitative analysis—reducing repetitive work and improving the reliability of my research.

5 Week3——Feature Extraction and Dimension Reduction

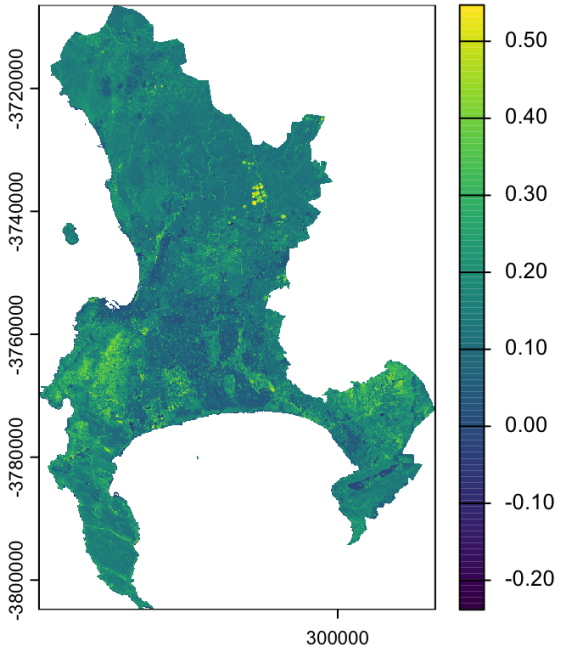
5.1 1. Summary

This learning task uses Landsat satellite imagery of Cape Town as its basis. Using the R programming language, we extracted the Normalized Difference Vegetation Index (NDVI) based on spectral features and calculated the Gray-Level Co-occurrence Matrix (GLCM) texture based on spatial features. Finally, we performed dimensional reduction and extracted key features from the multi-source fused data using Principal Component Analysis (PCA). This week's learning theme focuses on "Advanced Feature Extraction and Data Dimension Reduction." The primary objective of performing this series of preprocessing operations is to prepare the most concise and highest-quality feature variables for subsequent land cover classification (or machine learning models).

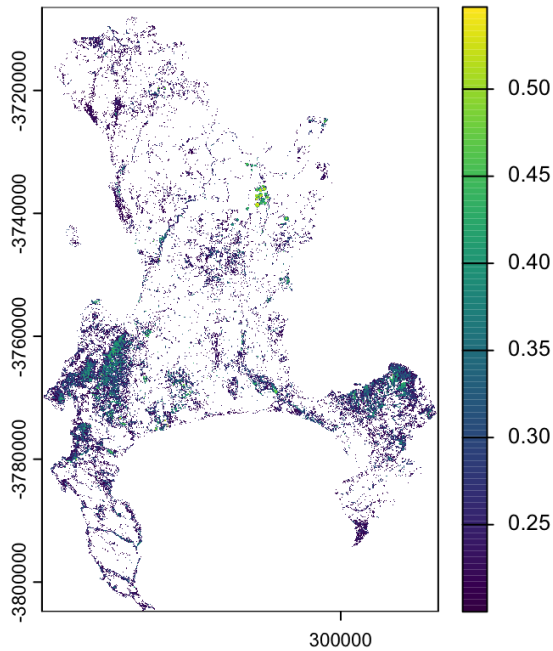
5.1.1 1.1 NDVI: Vegetation Feature Extraction

Based on the principle that different surface materials reflect light differently (healthy vegetation contains chlorophyll, which strongly absorbs red light, while the cellular structure of plants strongly reflects near-infrared light), the calculation is performed using the following formula:

$$NDVI = \frac{B5 - B4}{B5 + B4}$$



In the analysis of the Cape Town region, after calculating the NDVI for the entire area, we set a threshold of 0.2. By masking areas with values below this threshold, we successfully removed large areas of non-vegetation backgrounds, such as the ocean, dense urban structures, and bare soil. This process yielded a clean, valid vegetation cover map.

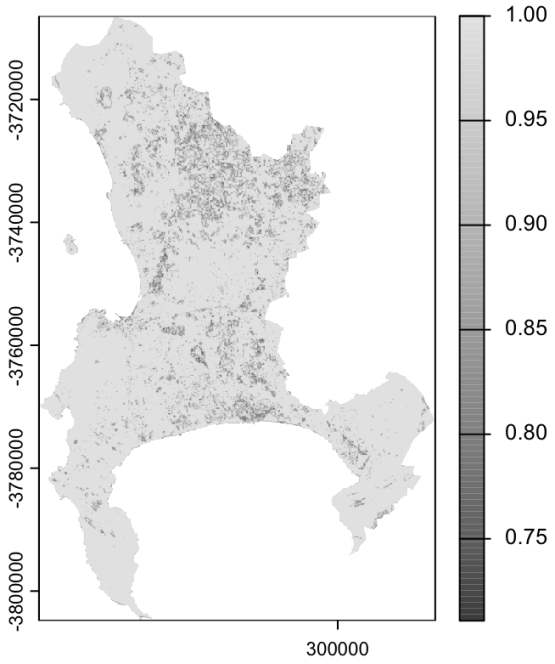


5.1.2 1.2 GLCM: Spatial Texture Feature Extraction

Satellite spectral bands mainly reflect surface color, but not spatial structure. Gray-Level Co-occurrence Matrix (GLCM) quantifies texture by analyzing grey-level variations between neighboring pixels, capturing surface smoothness or roughness.

The formula for converting DN to refractive index is $\rho = (DN \times 0.0000275) - 0.2$

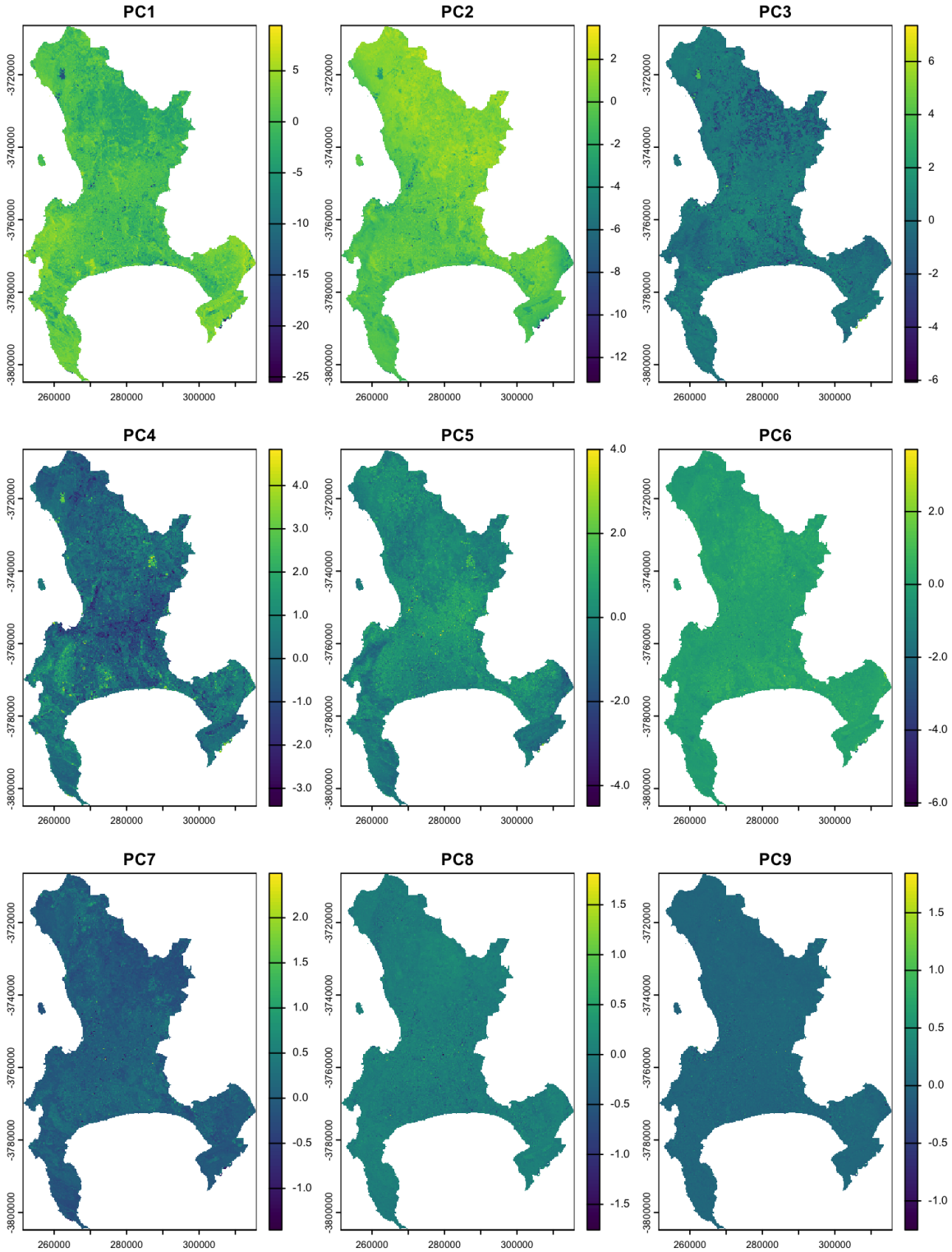
In this study, using Cape Town as an example, we calculated a 7×7 window homogeneity metric from the red band. The large areas of light gray and pure white in the image represent "high homogeneity," indicating that the surface in these areas is smooth, such as open water bodies, vegetation, or flat bare ground. The dark gray, densely spotted areas represent "low homogeneity," indicating that the texture of the adjacent surface is fragmented, corresponding to urban built-up areas, dense residential zones.



5.1.3 1.3 PCA: Multi-source Data Fusion and Dimension Reduction

There is typically a strong correlation between the various bands in multispectral imagery, leading to information redundancy and introducing a certain amount of noise. PCA uses an orthogonal transformation to convert these correlated layers into a set of mutually independent new variables, which are ranked by information content.

We applied PCA to eight spectral bands and one texture band from Cape Town. Most of the useful information is captured in the first few components, while noise and less important variation are pushed into later components. This helps reduce redundancy and provides clearer inputs for land cover classification.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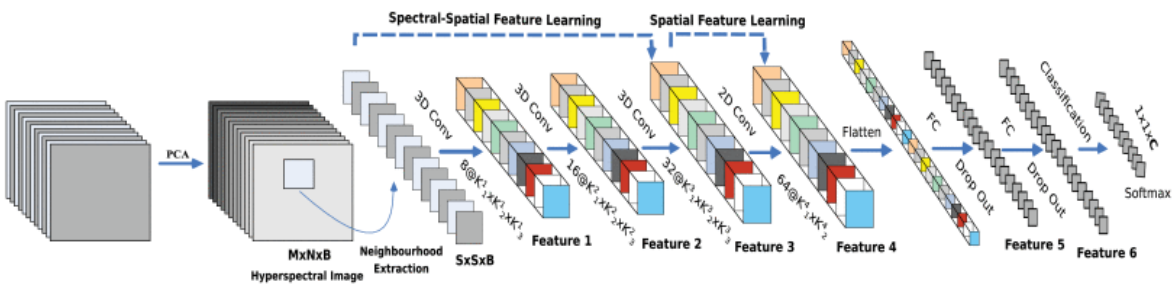
5.2 2. Application

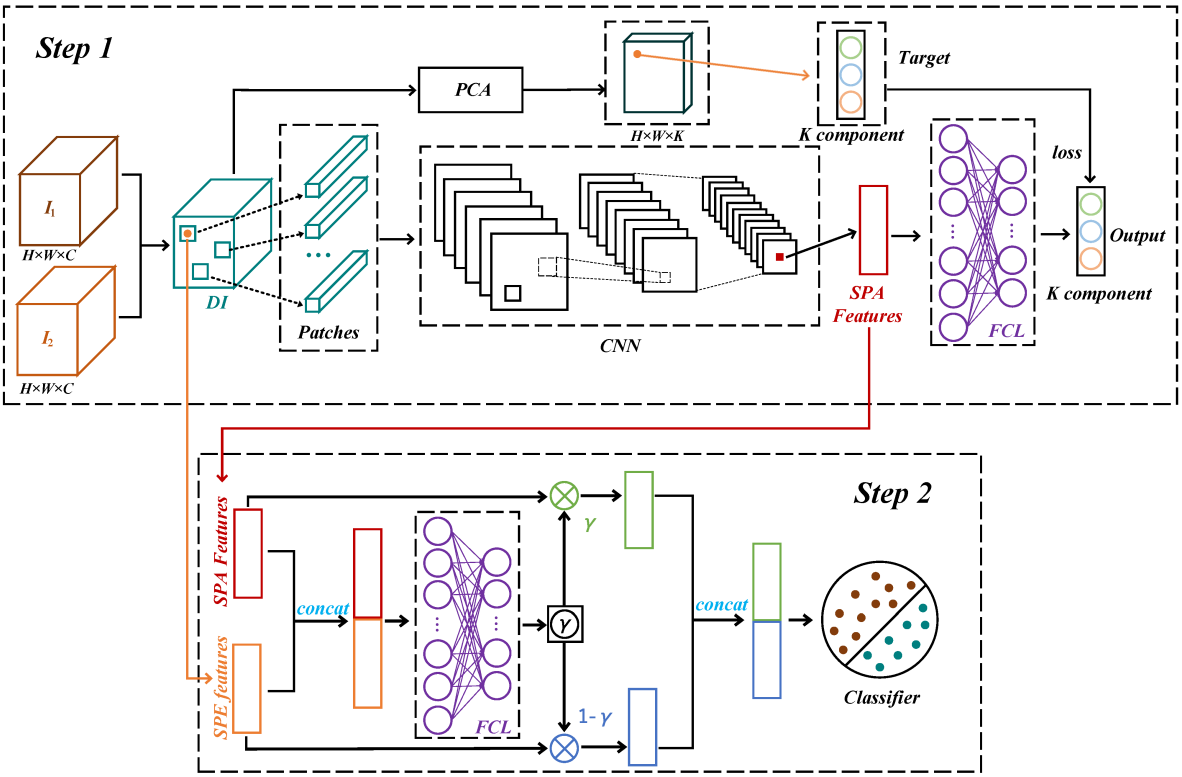
Principal Component Analysis (PCA), as a preprocessing step for dimensionality reduction, has been widely applied in remote sensing workflows. For example, (Zabalza et al. 2014) incorporated PCA into a broader feature extraction framework for hyperspectral and Synthetic Aperture Radar (SAR) data. PCA was primarily used to reduce data dimensionality and computational load.

Several recent studies have combined PCA with convolutional neural networks (CNNs). PCA is used to reduce spectral redundancy, while CNNs extract spatial features and perform classification. This approach significantly improves classification accuracy compared to traditional methods(Roy et al. 2020). Essentially, the capabilities of PCA itself have not been enhanced; rather, the introduction of CNNs compensates for PCA's limitations in representing spatial information.



In addition, some studies have developed more complex processing workflows by combining PCA with multi-stage feature extraction techniques, thereby capturing both global and local features of the data simultaneously. This suggests that in modern remote sensing analysis, PCA is more commonly used as a dimension-reduction tool rather than as a standalone feature extraction method.

- **PCA + Spatial-Spectral Fusion:** Fusing the global features extracted by PCA with the local features extracted by SSA significantly improves the accuracy of land cover classification(Wang et al. 2021).



- **PCA + Hybrid Manifold Learning:** Within PCA, apply LPP (Locally Linear Embedding) or t-SNE to preserve the local geometric proximity of the data(Cao et al. 2024).

5.3 3. Reflection

Through practical experience, I have found that PCA has some obvious drawbacks. For example, PCA combines the original features linearly into new principal components, which means the results lack intuitive physical or practical meaning. While the plot shows which directions exhibit the greatest variation, it is impossible to clearly determine which light waves are responsible for these changes. Furthermore, since PCA prioritizes retaining information with large variations, useful information may be lost in some practical tasks where features with small variations are often equally important for classification or recognition. Overall, while PCA effectively compresses information, its poor interpretability and insensitivity to nonlinearities and small-amplitude changes are limitations that must be weighed when using the method.

6 Week4——Policy

6.1 1. Introduction

Ahmedabad is one of India's typical high-temperature cities and has experienced frequent extreme heat waves over the past few decades. As urbanization progresses, the increase in impervious surfaces and the reduction in green spaces have further exacerbated the urban heat island effect. After analyzing data from 2002 to 2018, (A. Sharma et al. 2024) found a significant association between extreme heat events and all-cause mortality. Extreme heat is no longer merely a climatic phenomenon; it poses a substantial threat to public health.

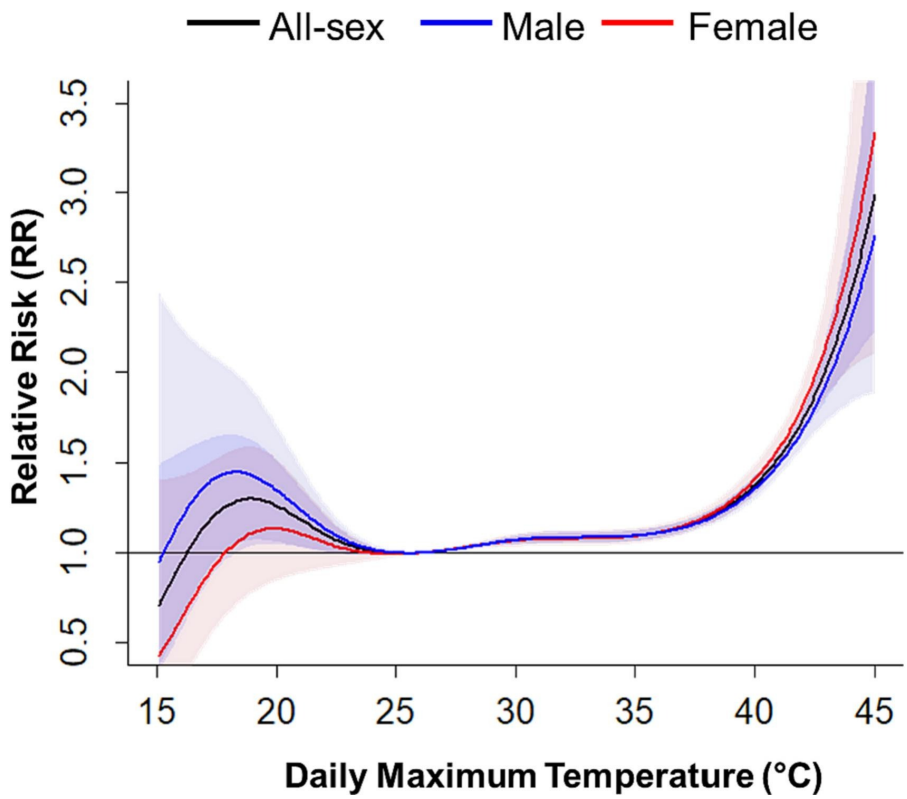


Figure 6.1: Association between Maximum Temperature and Mortality Risk. Resource: (A. Sharma et al. 2024).

6.2 2. Ahmedabad 2016 Heat Action Plan

Against this backdrop, Ahmedabad began to gradually explore systematic measures to address heatwaves, the most notable of which is the Ahmedabad Heat Action Plan. This policy is considered the first city-level heat-related health response plan in South Asia. It primarily aims to reduce the risks through early warning systems, public awareness campaigns, and interagency collaboration.

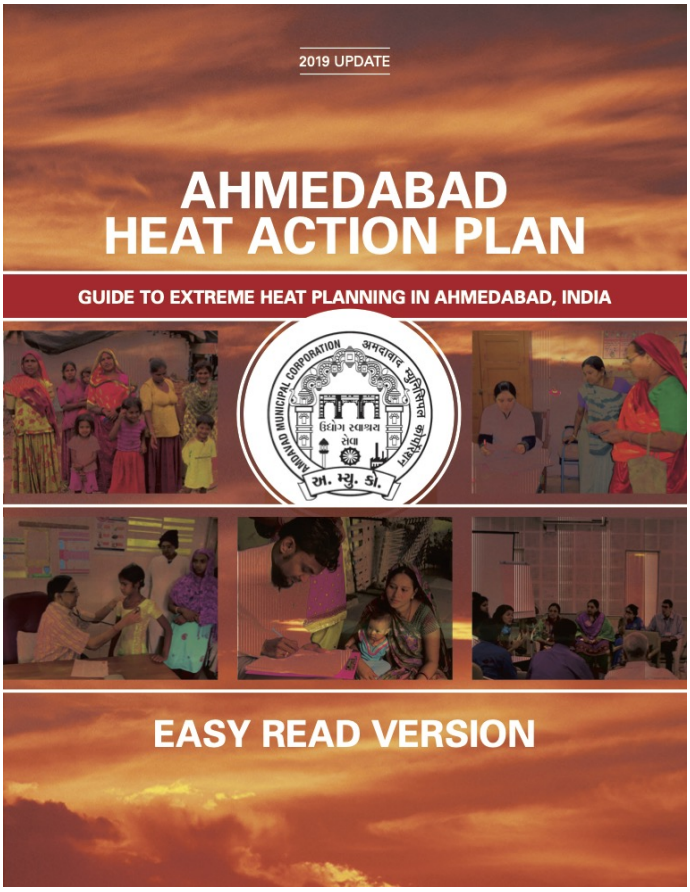


Figure 6.2: Ahmedabad Heat Action Plan 2019

As the policy was implemented, attention gradually turned to its effectiveness. Research findings indicate that following the policy's implementation, the risk of heat-related deaths has decreased (Hess et al. 2018).This result demonstrates that policy interventions can, to some extent, mitigate the impacts of extreme weather. This has led me to further consider whether disparities in heat risk persist across different areas within the city, and whether existing policies are capable of adequately identifying these disparities.

Current analytical methods remain primarily based on meteorological warnings and city-wide responses, with relatively limited attention paid to spatial variations within cities. Differences in vegetation cover, building density, and socioeconomic conditions across different areas all influence their sensitivity to high temperatures. Therefore, how to more precisely identify the distribution of heat risk within cities and provide support for more targeted interventions has become a topic worthy of further exploration.

6.3 3. Data

Given the limitations of existing Heat Action Plans in terms of spatial resolution. The incorporation of remote sensing data can help provide a better understanding of variations in heat exposure within cities. By analyzing multi-source remote sensing data, it is possible to identify which communities are more vulnerable to high temperatures. Furthermore, by combining this information with data on vegetation cover and water body distribution, we can pinpoint areas lacking natural factors that help mitigate heat stress.

The raw data for LST primarily comes from thermal infrared imagery (TIRS) captured by the Landsat-8 satellite. Surface emissivity is derived from the thermal infrared band, and surface temperature for each pixel is estimated through atmospheric correction and radiometric calibration. Some researchers have further utilized multispectral and high-resolution imagery to downscale LST to a 10-meter resolution, capturing spatial variations in the urban thermal environment(S. Sharma et al. 2023)

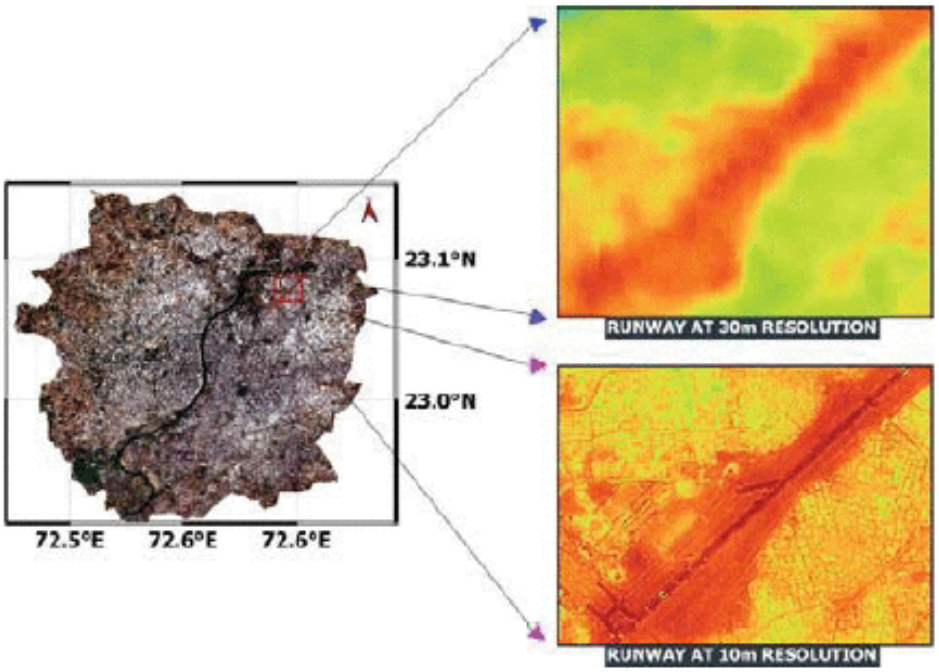


Figure 6.3: Visual comparison between 30m and 10m pixel size of ahmedabad airport runway (tarmac). Resource: (S. Sharma et al. 2023)

Raw NDVI data is derived from the visible and near-infrared bands of Landsat-8 or Sentinel-2. This index can be used to identify areas within cities with abundant or scarce green space, and when combined with LST data, it can be used to analyze the role of green space in mitigating high temperatures. Previous studies have also shown that vegetation and water bodies exhibit a negative correlation with surface temperature in the urban heat environment of Ahmedabad. Areas with abundant vegetation typically have lower temperatures(Mohammad, Goswami, and Bonafoni 2019).

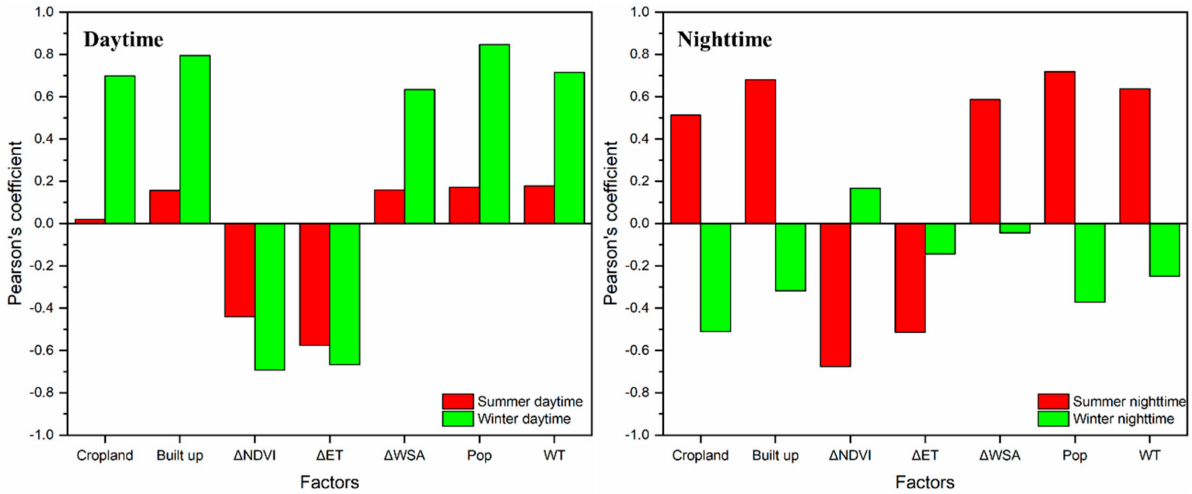


Figure 6.4: Pearson's correlation between SUHII (2003–2018) and different potential drivers during daytime and nighttime, in the summer and winter season. (Mohammad, Goswami, and Bonafoni 2019)

6.4 4. Application

As we mentioned earlier, we can now obtain Land Surface Temperature (LST) data with 10-meter resolution and detailed vegetation cover (NDVI) data. The most direct application of this is to transform these data into a high-resolution urban heat island diagnosis map. We no longer need to rely on citywide weather forecasts. Satellite imagery can provide precision down to the block or even street level, directly identifying which specific buildings and neighborhoods are currently in the furnace.

When a heat wave strikes, the government won't have to cast a wide net across the entire city. Where should cooling centers and temporary water stations be set up? In which areas should ambulances be on standby? Simply deploy resources directly to the red zones with the highest LST readings. At the same time, heat warnings no longer need to be broadcast citywide. Instead, by combining this data, targeted alerts can be sent to high-risk communities with dense populations and poor green space coverage, reminding residents to take precautions. Furthermore, by overlaying this data with population density or socioeconomic data. We can identify areas with high heat exposure and high vulnerability, which is critical for developing more equitable policies.

6.5 5. Reflections

The results derived from satellite image analysis are free from subjective bias. They enable decision-makers to quickly bypass the debate over "where resources are most needed" and move directly to the implementation phase of "how to carry out interventions." Furthermore, since this data is inexpensive to obtain and updated rapidly, it significantly reduces the burden on field staff responsible for data collection.

However, Satellites measure surface temperatures such as how hot a rooftop or asphalt road is. This does not equate to the air temperature or indoor temperature that residents actually experience. An area with extremely high LST and low NDVI could be a large, modern shopping mall with central air conditioning. Conversely, an urban village that appears to be surrounded by greenery may actually consist of shantytowns built with corrugated iron roofs and extremely poor ventilation, where residents are enduring deadly indoor heat. Relying solely on "red zones" on satellite imagery to allocate relief supplies can easily lead to a severe misallocation of resources.

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